

WESTERN REFRIGERATION PVT. LTD.

Functional Module Specification

Product: WRL Portal

Version: 1.0 draft

Date: 2026-09-02

Production: <https://wrl-dashboard.vercel.app>

Internal — Handover Documentation

Status: Ready for IT / functional review (repo-generated from module READMEs + architecture). Describes what each module does today — not a stakeholder-signed requirements doc. Pair with [BRD WRL Portal.md](#) for business context.

Module-by-module functional reference. Technical implementation: `03-Technical/ARCHITECTURE.md`.

mis — MIS Reports

Field	Detail
Purpose	Primary ops reporting: Summary, Call Register, Accounts, BD-MIS, Deployment Completion, client import
Users / roles	<code>page_mis_reports</code> + tab permissions (<code>tab_mis_summary</code> , <code>tab_mis_register</code> , etc.)
Screens	<code>/report</code> (tabs under one page)
Key actions	Filter by date/branch/franchisee/status (role baseline on load; Clear all on register); export CSV/Excel; open TRN call drawer; upload client import files
Data sources	Postgres hot tables + CRM SQL for register/detail; client import in Postgres
Scheduled jobs	None (consumes read-model sync)
RBAC	Page <code>mis_reports</code> ; tabs per MIS_TABS in RBAC matrix

mis-email — Mail & Alerts

Field	Detail
Purpose	MIS digests, routing rules, major-repair alerts, cancelled-call digests; unified email send
Users / roles	<code>page_mis_email_settings</code> or OR-gate: <code>manage_users</code> , <code>manage_roles</code> , <code>view_all_offices</code> , legacy mail page perms

Field	Detail
Screens	<code>/admin/mis-email-settings</code> (tabs: org, routing, major repair, cancelled, VPS cron, subcontractor)
Key actions	Configure outbound gate, routing rules, recipient lists; test send; view send tracker
Data sources	Postgres prefs/routing/recipient tables; report data via mis module services
Scheduled jobs	MIS digest <code>*/15</code> via <code>mail-scheduler.sh</code> ; cancelled digest 16:00 IST (<code>evening-ops-sequencer.sh</code>); major-repair in sync worker (not cron)
RBAC	<code>page_mis_email_settings</code> , <code>page_major_repair_alerts</code> , <code>page_cancelled_call_alerts</code> (legacy paths)

subcontractor-stock — Subcontractor stock

Field	Detail
Purpose	SAP inbox → CRM stock reconcile → discrepancy Excel → plant email
Users / roles	<code>page_mis_email_settings</code> (tab under Mail & Alerts)
Screens	<code>/admin/mis-email-settings?tab=subcontractor</code>
Key actions	Sync inbox, run reconciliation, send emails, manage skip rules and recipients
Data sources	SAP attachments (inbox), CRM stock query, Postgres config/history
Scheduled jobs	VPS CLI: inbox poll, auto-reconcile, IST send time
RBAC	Same as mis-email settings page

cancelled-calls — Cancelled Calls

Field	Detail
Purpose	Cancelled call register from Postgres; CSV export; feeds cancelled digest

Field	Detail
Users / roles	page_cancelled_calls
Screens	/report/cancelled-calls
Key actions	Filter, paginate, export CSV
Data sources	Postgres read-model (sync from CRM)
Scheduled jobs	Sync via read-model worker; digest via mis-email
RBAC	page_cancelled_calls

athena-reconciliation — Failed Calls (Athena API)

Field	Detail
Purpose	CRM Athena ingestion failures matched to call register
Users / roles	page_athena_reconciliation
Screens	/report/athena-reconciliation
Key actions	View reconciliation rows; sync trigger via read-model
Data sources	Postgres + CRM fetch for Athena failures
Scheduled jobs	Read-model sync pipeline
RBAC	page_athena_reconciliation

attendance — Service call activity

Field	Detail
Purpose	Technician activity report (admin)
Users / roles	manage_users (admin sync/attendance paths)

Field	Detail
Screens	/admin/attendance
Key actions	Filter activity; CSV export
Data sources	Postgres attendance read-model
Scheduled jobs	Attendance sync in read-model worker
RBAC	manage_users

distribution — Call Distribution

Field	Detail
Purpose	Franchisee map, idle assignees, distribution KPIs
Users / roles	page_call_distribution
Screens	/report/distribution
Key actions	Filter; view map/table; CSV export
Data sources	Postgres/CRM via shared MIS filters
Scheduled jobs	None
RBAC	page_call_distribution

arcp-claims — ARCP Claims

Field	Detail
Purpose	ARCP claims register and detail export
Users / roles	page_arcp_claims
Screens	/report/arcp-claims

Field	Detail
Key actions	Browse register; export
Data sources	CRM / Postgres per module queries
Scheduled jobs	None
RBAC	page_arcp_claims

serial-audit — Serial Wise History

Field	Detail
Purpose	Repeat serial complaints and repair audit
Users / roles	page_serial_audit
Screens	/report/serial-audit
Key actions	Search serial; expand history; CSV export
Data sources	CRM + hot read-model
Scheduled jobs	None
RBAC	page_serial_audit

location-audit — Location Audit

Field	Detail
Purpose	Technician visit location vs customer install address
Users / roles	page_location_audit
Screens	/report/location-audit
Key actions	Flag mismatches; filter; export

Field	Detail
Data sources	CRM geo + install data
Scheduled jobs	None
RBAC	page_location_audit

warranty-master — Warranty Master

Field	Detail
Purpose	Active machines by customer, group, warranty period
Users / roles	page_warranty_master
Screens	/report/warranty-master
Key actions	Filter; browse; export
Data sources	CRM
Scheduled jobs	None
RBAC	page_warranty_master

users — User Management

Field	Detail
Purpose	Create and edit portal users
Users / roles	manage_users
Screens	/admin/users
Key actions	CRUD users; assign role; office scope; enable MIS email
Data sources	Supabase auth + app_users Postgres

Field	Detail
Scheduled jobs	None
RBAC	<code>manage_users</code>

roles — Roles & Access

Field	Detail
Purpose	Define roles and page/tab/capability permissions
Users / roles	<code>manage_roles</code>
Screens	<code>/admin/roles</code>
Key actions	Edit role permission matrix; hierarchical MIS tabs
Data sources	<code>app_roles</code> , <code>app_role_permissions</code>
Scheduled jobs	None
RBAC	<code>manage_roles</code>

auth — Authentication

Field	Detail
Purpose	Login, session, password flows
Users / roles	All users
Screens	<code>/login</code> , <code>/profile</code>
Key actions	Sign in/out; update profile (name, avatar, theme, password); MIS email prefs if granted
Data sources	Supabase
Scheduled jobs	None

Field	Detail
RBAC	Session-based; landing page from first permitted route

sync — Read-model sync (admin)

Field	Detail
Purpose	Sync status dashboard (admin)
Users / roles	<code>manage_users</code> for <code>/admin/sync</code> view
Screens	<code>/admin/sync</code>
Key actions	View worker status; refresh status
Data sources	VPS workers + Postgres read-model tables
Scheduled jobs	systemd sync workers on VPS (see SYNC_ENTRY_POINTS)
RBAC	<code>manage_users</code> . Manual CRM sync → yesterday is on <code>/report</code> Call Register tab — super_admin only (<code>/api/admin/calls-hot-sync</code>)

security-audit — Activity log

Field	Detail
Purpose	Privileged action audit trail
Users / roles	<code>super_admin</code> capability
Screens	<code>/admin/security-audit</code> (Activity Log in sidebar Admin section)
Key actions	Search audit events
Data sources	Postgres audit log
Scheduled jobs	None

Field	Detail
RBAC	<code>super_admin</code>

performance — Performance Insights

Field	Detail
Purpose	Client performance metrics and diagnostics
Users / roles	<code>page_performance_insights</code>
Screens	<code>/admin/performance-insights</code>
Key actions	View metrics; diagnostics
Data sources	Client-side performance collection
Scheduled jobs	None
RBAC	<code>page_performance_insights</code>

calls — Call detail API

Field	Detail
Purpose	Live CRM call drawer data, offices list, comments
Users / roles	<code>tab_mis_register</code> (Call Register tab) + office scope
Screens	Call drawer from Call Register (not standalone page)
Key actions	Fetch call detail; post comments; flags
Data sources	CRM live + Postgres flags/comments
Scheduled jobs	None
RBAC	<code>resolveReportSecurity</code> with <code>pageId: mis_reports</code> , <code>tabId: register</code>

Cross-cutting

Concern	Where documented
RBAC catalog	04-RBAC/ , src/lib/auth/rbac-catalog.ts
Read-model workers	05-Operations/SYNC_ENTRY_POINTS.md
Mail cron	05-Operations/MAIL_SCHEDULE.md
Prod env	05-Operations/PROD_READ_SOURCE.md